

Experiment 1: Case Study on KL ERP System

Name: G. Venkata Varun

Rollno: 2420030187

Course: Adaptive Software Engineering

Section: 6

Problem Statement

Choose a real-world case study such as the **KL ERP System**, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

1. Functional Requirements

Student

- Register/Login using university credentials
- View profile information
- Access course registration
- View attendance
- Check examination schedule
- View grades and academic progress
- Download fee receipts
- View timetable
- Submit service requests

Faculty

- Login securely
- Manage attendance

- Upload marks
- View assigned courses
- Share study materials
- View student information

Administrator

- Manage student accounts
 - Manage faculty accounts
 - Manage courses
 - Update academic calendar
 - Generate reports
 - Manage fee records
 - Maintain ERP database
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2. User Stories

User ID	User Story	Priority
US-01	As a student, I want to login securely so that I can access my academic information.	High
US-02	As a student, I want to view my attendance so that I can monitor my eligibility.	High
US-03	As a student, I want to register for courses online.	High
US-04	As a student, I want to view my grades after examinations.	High
US-05	As a student, I want to download my fee receipt.	Medium
US-06	As a faculty member, I want to upload student marks efficiently.	High

User ID	User Story	Priority
US-07	As a faculty member, I want to record attendance digitally.	High
US-08	As an administrator, I want to manage student accounts.	High
US-09	As an administrator, I want to update the academic calendar.	Medium
US-10	As an administrator, I want to generate academic and administrative reports.	High
US-11	As an administrator, I want to manage course offerings.	Medium
US-12	As a student, I want to submit service requests through ERP.	Medium

3. Selected Software Development Process Model

Agile Scrum

Reason for Selection

- University requirements frequently change.
- Easy to accommodate academic policy updates.
- Continuous stakeholder feedback.
- Incremental module delivery.
- Easier maintenance.
- Suitable for large web-based ERP systems.

Sprint Plan

Sprint	Features
Sprint 1	Student Login & Authentication
Sprint 2	Student Dashboard
Sprint 3	Attendance & Timetable
Sprint 4	Course Registration
Sprint 5	Examination & Results
Sprint 6	Faculty Module
Sprint 7	Admin Dashboard
Sprint 8	Testing & Deployment

4. Programming Language Selection

Backend

- Java
- Spring Boot

Reasons

- Enterprise-grade framework
- Highly secure
- Excellent database connectivity
- REST API support
- Scalable architecture

Frontend

- HTML5
- CSS3
- JavaScript

- React.js

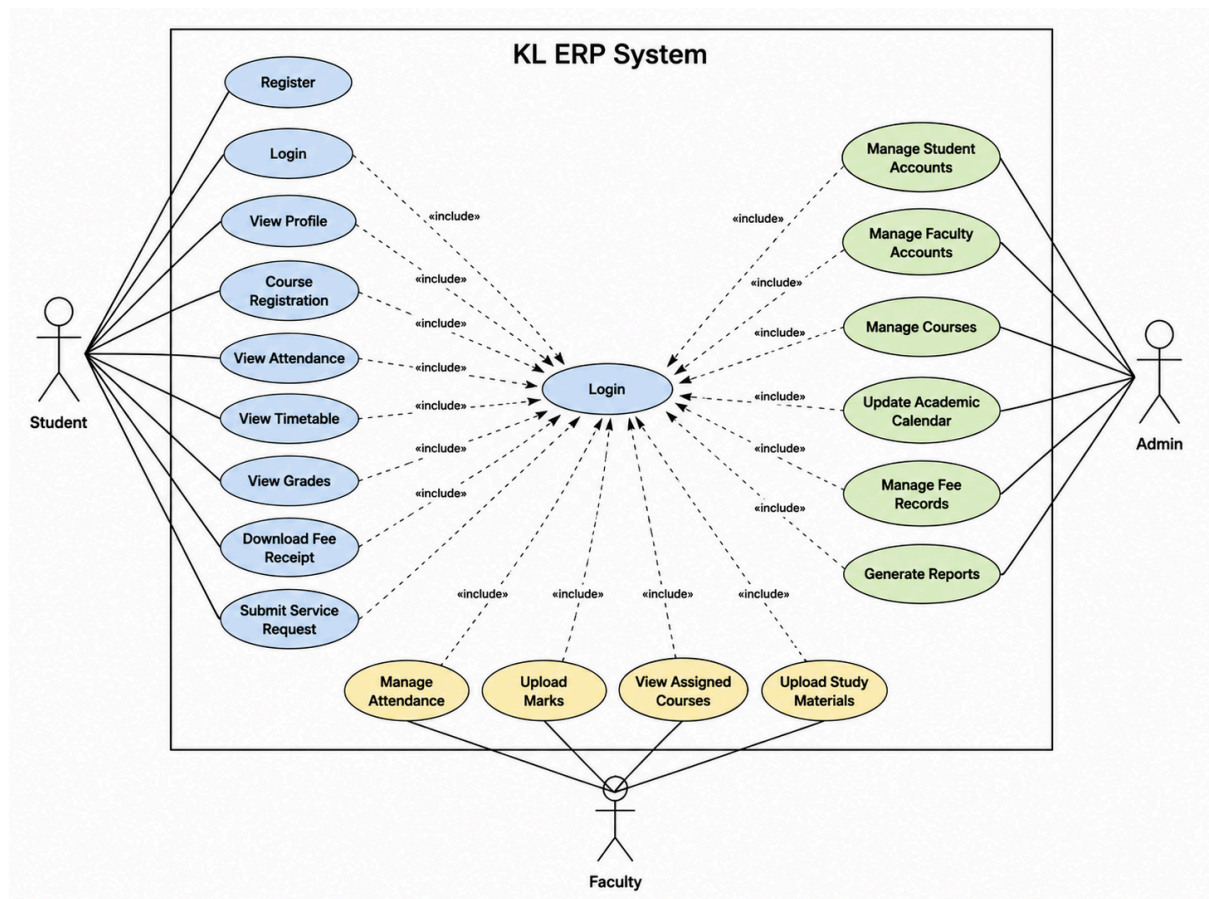
Database

- MySQL

Other Technologies

- REST API
- JWT Authentication
- Git
- Docker
- AWS Cloud

5. System Architecture



6. Implementation Strategy

Phase 1 – Requirement Analysis

- Gather requirements from students, faculty, and administrators.
 - Prepare Software Requirement Specification (SRS).
 - Create use case diagrams.
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Phase 2 – System Design

- Design database schema.
 - Design user interface wireframes.
 - Create UML diagrams.
 - Develop system architecture.
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Phase 3 – Development

- Develop an authentication module.
 - Implement a student dashboard.
 - Implement attendance management.
 - Develop course registration module.
 - Build a faculty portal.
 - Build administrator dashboard.
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Phase 4 – Testing

Unit Testing

- Login
- Attendance
- Grade viewing
- Course registration

Integration Testing

- Attendance + Faculty Module
- Course Registration + Database
- Results + Student Dashboard

System Testing

- Complete ERP application testing.

User Acceptance Testing (UAT)

- Verify academic and administrative requirements.
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Phase 5 – Deployment

- Docker Containers
 - AWS EC2
 - MySQL Database
 - Nginx Web Server
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7. Evaluation Strategy

Evaluation Parameter	Method
Functional Accuracy	Requirement verification
Performance	Load testing using Apache JMeter
Security	Authentication and penetration testing
Reliability	Continuous monitoring
Usability	Student and faculty feedback
Scalability	Concurrent user testing
Maintainability	Code review and documentation

8. Expected Outcomes

- Secure authentication for students, faculty, and administrators.
- Efficient attendance management.
- Easy online course registration.
- Accurate grade management.
- Reliable fee management.
- Improved academic administration.
- Scalable ERP architecture for future enhancements.